

101. iAP2 Control Session Messages

101.1 Accessory Authentication

Requirements and usage for these messages may be found in [Accessory Authentication](#) (page 276).

101.1.1 RequestAuthenticationCertificate

Source	ID
Device	0xAA00

Table 101-1 RequestAuthenticationCertificate message parameters

Name	ID	Type	#	Notes
RequestAuthenticationCertificateSerialNumber	0	none	0/1	

101.1.2 AuthenticationCertificate

Source	ID
Accessory	0xAA01

Table 101-2 AuthenticationCertificate message parameters

Name	ID	Type	#	Notes
AuthenticationCertificate	0	blob	1	Accessory's X.509 certificate.

101.1.3 RequestAuthenticationChallengeResponse

Source	ID
Device	0xAA02

Table 101-3 RequestAuthenticationChallengeResponse message parameters

Name	ID	Type	#	Notes
AuthenticationChallenge	0	blob	1	Random #.

101.1.4 AuthenticationResponse

Source	ID
Accessory	0xAA03

Table 101-4 AuthenticationResponse message parameters

Name	ID	Type	#	Notes
AuthenticationResponse	0	blob	1	Computed challenge response.

101.1.5 AuthenticationFailed

Source	ID
Device	0xAA04

Table 101-5 AuthenticationFailed message parameters

Name	ID	Type	#	Notes
This message has no parameters.				

101.1.6 AuthenticationSucceeded

Source	ID
Device	0xAA05

Table 101-6 AuthenticationSucceeded message parameters

Name	ID	Type	#	Notes
This message has no parameters.				

101.1.7 AccessoryAuthenticationSerialNumber

Source	ID
Accessory	0xAA06

Table 101-7 AccessoryAuthenticationSerialNumber message parameters

Name	ID	Type	#	Notes
AuthenticationSerialNumber	0	blob	1	Accessory's X.509 certificate serial number as binary data.

101.2 Accessory Identification

Requirements and usage for these messages may be found in [Accessory Identification](#) (page 281).

101.2.1 StartIdentification

Source	ID
Device	0x1D00

Table 101-8 StartIdentification message parameters

Name	ID	Type	#	Notes
This message has no parameters.				

101.2.2 IdentificationInformation

Source	ID
Accessory	0x1D01

Table 101-9 IdentificationInformation message parameters

Name	ID	Type	#	Notes
Name	0	utf8	1	Shall match the accessory's markings and packaging. An empty string is not allowed.
ModelIdentifier	1	utf8	1	Shall match the accessory's markings and packaging. An empty string is not allowed.

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101.2 Accessory Identification

Name	ID	Type	#	Notes
Manufacturer	2	utf8	1	Shall match the accessory's markings and packaging. An empty string is not allowed.
SerialNumber	3	utf8	1	Shall match the accessory's markings and packaging. An empty string is not allowed.
FirmwareVersion	4	utf8	1	Shall uniquely reflect the current revision of the accessory's firmware. An empty string is not allowed.
HardwareVersion	5	utf8	1	Shall uniquely reflect the current revision of the accessory's hardware. An empty string is not allowed.
MessagesSentByAccessory	6	uint16[0+]	1	The exhaustive set of unique messages that this accessory will send. This set is expressed as an array of uint16 message identifiers.
MessagesReceivedFromDevice	7	uint16[0+]	1	The exhaustive set of unique messages that this accessory expects to receive. This set is expressed as an array of uint16 message identifiers.
PowerProvidingCapability	8	enum	1	See PowerProvidingCapability (page 874). Shall be None if the accessory does not provide power to the device or does not use iAP2 to declare its power providing capability. See Device Power (Lightning) (page 328).
MaximumCurrentDrawnFromDevice	9	uint16	1	Maximum current drawn by accessory from Accessory Power pin in mA. Shall be 0 if accessory does not draw power from the device.
SupportedExternalAccessoryProtocol	10	group	0+	See ExternalAccessoryProtocol (page 874).
AppMatchTeamID	11	utf8	0/1	See App Match (page 301).
CurrentLanguage	12	utf8	1	The accessory's current active language setting. Shall be one of the supported languages.
SupportedLanguage	13	utf8	1+	A language supported by the accessory. Use the ISO 639-1 designation unless it is not available, in which case use the ISO 639-2 designation. See https://www.loc.gov/standards/iso639-2/php/English_list.php for a complete list of ISO 639-1 and ISO 639-2 codes.
UARTTransportComponent	14	group	0/1	See UARTTransportComponent (page 876).
USBDeviceTransportComponent	15	group	0/1	See USBDeviceTransportComponent (page 875).
USBHostTransportComponent	16	group	0/1	See USBHostTransportComponent (page 875).
BluetoothTransportComponent	17	group	0+	See BluetoothTransportComponent (page 876).
iAP2HIDComponent	18	group	0+	See iAP2HIDComponent (page 876).
LocationInformationComponent	22	group	0/1	See LocationInformationComponent (page 876).

Name	ID	Type	#	Notes
USBHostHIDComponent	23	group	0+	See USBHostHIDComponent (page 877).
BluetoothHIDComponent	29	group	0/1	See BluetoothHIDComponent (page 877).
ProductPlanUID	34	utf8	1	Shall match the accessory's 16 character Product Plan UID. The UID is different from the Product Plan ID. An empty string is not allowed.

Table 101-10 PowerProvidingCapability enum

Value	Description
0	None
1	Lightning Receptacle (C37) Passthrough
2	Advanced

Table 101-11 ExternalAccessoryProtocol parameter group

Name	ID	Type	#	Notes
ExternalAccessoryProtocolIdentifier	0	uint8	1	All ExternalAccessoryProtocol identifiers shall be unique.
ExternalAccessoryProtocolName	1	utf8	1	
ExternalAccessoryProtocolMatchAction	2	enum	1	See MatchAction (page 874) and App Match (page 301).
NativeTransportComponentIdentifier	3	uint16	0/1	Shall refer to the TransportComponentIdentifier of a declared USBHostTransportComponent (page 875).

Table 101-12 MatchAction enum

Value	Description
0	No Action. The device will not prompt the user to find a matching app, and there will not be a <i>Find App For This Accessory</i> button in <i>Settings > General > About > 'Accessory Name'</i> .
1	Optional Action. The device may prompt the user to find a matching app, and there will be a <i>Find App For This Accessory</i> button in <i>Settings > General > About > 'Accessory Name'</i> . See App Match (page 301) for more information.
2	No Alert. The device will not prompt the user to find a matching app, but there will be a <i>Find App For This Accessory</i> button in <i>Settings > General > About > 'Accessory Name'</i> . See App Match (page 301) for more information.
3	No Communication Protocol. The protocol is not intended for communication and does not require StartExternalAccessoryProtocolSession (page 896) or StopExternalAccessoryProtocolSession (page

Value	Description
	896). The device may prompt the user to find a matching app, and there will be a <i>Find App For This Accessory</i> button in <i>Settings > General > About > 'Accessory Name'</i> . See App Match (page 301) for more information.

Table 101-13 USBDeviceTransportComponent parameter group

Name	ID	Type	#	Notes
TransportComponentIdentifier	0	uint16	1	All TransportComponent identifiers shall be unique.
TransportComponentName	1	utf8	1	
TransportSupportsIAP2Connection	2	none	0/1	
USBDeviceSupportedAudioSampleRate	3	enum	0+	See USBDeviceModeAudioSampleRate (page 875).

Table 101-14 USBDeviceModeAudioSampleRate enum

Value	Description
0	8000 Hz
1	11025 Hz
2	12000 Hz
3	16000 Hz
4	22050 Hz
5	24000 Hz
6	32000 Hz
7	44100 Hz
8	48000 Hz

Table 101-15 USBHostTransportComponent parameter group

Name	ID	Type	#	Notes
TransportComponentIdentifier	0	uint16	1	All TransportComponent identifiers shall be unique.
TransportComponentName	1	utf8	1	
TransportSupportsIAP2Connection	2	none	0/1	

Table 101-16 UARTTransportComponent parameter group

Name	ID	Type	#	Notes
TransportComponentIdentifier	0	uint16	1	All TransportComponent identifiers shall be unique.
TransportComponentName	1	utf8	1	
TransportSupportsAP2Connection	2	none	0/1	

Table 101-17 BluetoothTransportComponent parameter group

Name	ID	Type	#	Notes
TransportComponentIdentifier	0	uint16	1	All TransportComponent identifiers shall be unique.
TransportComponentName	1	utf8	1	
TransportSupportsAP2Connection	2	none	0/1	
BluetoothTransportMediaAccessControlAddress	3	uint8[6]	1	The accessory's Bluetooth MAC address, a valid 6-byte IEEE EUI-48 identifier.

Table 101-18 iAP2HIDComponent parameter group

Name	ID	Type	#	Notes
HIDComponentIdentifier	0	uint16	1	All HIDComponentIdentifiers shall be unique.
HIDComponentName	1	utf8	1	
HIDComponentFunction	2	enum	1	See HIDComponentFunction (page 877).

Table 101-19 LocationInformationComponent parameter group

Name	ID	Type	#	Notes
Identifier	0	uint16	1	All Identifiers shall be unique.
Name	1	utf8	1	
GlobalPositioningSystemFixData	17	none	0/1	If present, the accessory is capable of generating NMEA GPGGA sentences.
RecommendedMinimumSpecificGPSTransitData	18	none	0/1	If present, the accessory is capable of generating NMEA GPRMC sentences.
VehicleSpeedData	20	none	0/1	If present, the accessory is capable of generating NMEA PASCD sentences.

Table 101-20 USBHostHIDComponent parameter group

Name	ID	Type	#	Notes
HIDComponentIdentifier	0	uint16	1	All HIDComponentIdentifiers shall be unique.
HIDComponentName	1	utf8	1	
HIDComponentFunction	2	enum	1	See HIDComponentFunction (page 877).
USBHostTransportComponentIdentifier	3	uint16	1	Shall refer to a USBHostTransportComponent (page 875).
USBHostTransportInterfaceNumber	4	uint16	1	Shall match the accessory's corresponding USB device interface descriptor. If more than one USBHostHIDComponent is present, the accessory shall present multiple USB HID interfaces with unique interface numbers.

Table 101-21 HIDComponentFunction enum

Value	Description
0	Keyboard
1	Media Playback Remote
2	AssistiveTouch
3	Reserved
4	Gamepad (device-holding)
5	Reserved
6	Gamepad
7	Assistive Switch Control
8	Headset
10	Braille Display/Keyboard

Table 101-22 BluetoothHIDComponent parameter group

Name	ID	Type	#	Notes
HIDComponentIdentifier	0	uint16	1	All HIDComponentIdentifiers shall be unique.
HIDComponentName	1	utf8	1	
HIDComponentFunction	2	enum	1	See HIDComponentFunction (page 877).
BluetoothTransportComponentIdentifier	3	uint16	1	Shall refer to a BluetoothTransportComponent (page 876).

101.2.3 IdentificationAccepted

Source	ID
Device	0x1D02

Table 101-23 IdentificationAccepted message parameters

Name	ID	Type	#	Notes
This message has no parameters.				

101.2.4 IdentificationRejected

Source	ID
Device	0x1D03

Table 101-24 IdentificationRejected message parameters

Name	ID	Type	#	Notes
Name	0	none	0/1	
ModelIdentifier	1	none	0/1	
Manufacturer	2	none	0/1	
SerialNumber	3	none	0/1	
FirmwareVersion	4	none	0/1	
HardwareVersion	5	none	0/1	
MessagesSentByAccessory	6	uint16[0+]	0/1	The set of unsupported messages sent by the accessory. This set is expressed as an array of uint16 message identifiers.
MessagesReceivedFromDevice	7	uint16[0+]	0/1	The set of unsupported messages received from the device. This set is expressed as an array of uint16 message identifiers.
PowerProvidingCapability	8	none	0/1	
MaximumCurrentDrawnFromDevice	9	none	0/1	
SupportedExternalAccessoryProtocol	10	none	0/1	One or more of the External Accessory Protocols is not supported by the device.
AppMatchTeamID	11	none	0/1	
CurrentLanguage	12	none	0/1	
SupportedLanguage	13	none	0/1	One or more of the identified languages is not supported by the device.
UARTTransportComponent	14	none	0/1	

Name	ID	Type	#	Notes
USBDeviceTransportComponent	15	none	0/1	
USBHostTransportComponent	16	none	0/1	
BluetoothTransportComponent	17	none	0/1	One or more of the identified Bluetooth Transport components is not supported by the device.
iAP2HIDComponent	18	none	0+	One or more of the identified HID components is not supported by the device.
LocationInformationComponent	22	none	0/1	One or more of the identified Location Information components is not supported by the device.
USBHostHIDComponent	23	none	0+	One or more of the identified HID components is not supported by the device.
BluetoothHIDComponent	29	none	0/1	One or more of the identified HID components is not supported by the device.
ProductPlanUID	34	none	0/1	

101.2.5 CancelIdentification

Source	ID
Accessory	0x1D05

Table CancelIdentification message parameters
101-25

Name	ID	Type	#	Notes
This message has no parameters.				

101.2.6 IdentificationInformationUpdate

Source	ID
Accessory	0x1D06

Table IdentificationInformationUpdate message parameters
101-26

Name	ID	Type	#	Notes
Name	0	utf8	1	
ModelIdentifier	1	utf8	1	
Manufacturer	2	utf8	1	
SerialNumber	3	utf8	1	

Name	ID	Type	#	Notes
FirmwareVersion	4	utf8	1	
HardwareVersion	5	utf8	1	
CurrentLanguage	6	utf8	1	

101.3 App Launch

Requirements and usage for these messages may be found in [App Launch](#) (page 299).

101.3.1 RequestAppLaunch

Source	ID
Accessory	0xEA02

Table RequestAppLaunch message parameters
101-27

Name	ID	Type	#	Notes
AppBundleID	0	utf8	1	Uniform Type Identifier (UTI) in reverse-DNS format (for example, com.Ajax.hello).
AppLaunchMethod	1	enum	0/1	See AppLaunchMethod (page 880).

Table AppLaunchMethod enum
101-28

Value	Description
0	Launch with user alert (default)
1	Launch without user alert

101.4 App Discovery

Requirements and usage for these messages may be found in [App Discovery](#) (page 298).

101.4.1 StartAppDiscoveryUpdates

Source	ID
Accessory	0xAD00

Table 101-29 StartAppDiscoveryUpdates message parameters

Name	ID	Type	#	Notes
ExternalAccessoryAppCategories	3	group	0/1	See ExternalAccessoryAppCategories (page 881).
ExternalAccessoryAppListMax	4	uint16	0/1	Maximum number of app list entries to send.

Table 101-30 ExternalAccessoryAppCategories parameter group

Name	ID	Type	#	Notes
AllExternalAccessoryApps	0	none	1	Request the list of apps that support the accessory's registered protocols.

101.4.2 AppDiscoveryUpdate

Source	ID
Device	0xAD01

Table 101-31 AppDiscoveryUpdate message parameters

Name	ID	Type	#	Notes
ExternalAccessoryAppListAvailable	3	enum	0/1	See AppListAvailable (page 881).
ExternalAccessoryAppList	4	group	0+	See AppList (page 881).
ExternalAccessoryAppListCount	5	uint16	0/1	Number of entries in ExternalAccessoryAppList.

Table 101-32 AppListAvailable enum

Value	Description
1	Unavailable
2	Available

Table 101-33 AppList parameter group

Name	ID	Type	#	Notes
Index	0	uint16	0/1	
AppBundleID	1	utf8	0/1	Uniform Type Identifier (UTI) in reverse-DNS format (for example, com.Ajax.hello).

Name	ID	Type	#	Notes
AppDisplayName	2	utf8	0/1	
AppCategoryID	3	uint16	0/1	ID from ExternalAccessoryAppCategories (page 881).

101.4.3 StopAppDiscoveryUpdates

Source	ID
Accessory	0xAD02

Table StopAppDiscoveryUpdates message parameters
101-34

Name	ID	Type	#	Notes
This message has no parameters.				

101.5 AssistiveTouch

Requirements and usage for these messages may be found in [AssistiveTouch](#) (page 307).

101.5.1 StartAssistiveTouch

Source	ID
Accessory	0x5400

Table StartAssistiveTouch message parameters
101-35

Name	ID	Type	#	Notes
This message has no parameters.				

101.5.2 StopAssistiveTouch

Source	ID
Accessory	0x5401

Table StopAssistiveTouch message parameters**101-36**

Name	ID	Type	#	Notes
This message has no parameters.				

101.5.3 StartAssistiveTouchInformation

Source	ID
Accessory	0x5402

Table StartAssistiveTouchInformation message parameters**101-37**

Name	ID	Type	#	Notes
This message has no parameters.				

101.5.4 AssistiveTouchInformation

Source	ID
Device	0x5403

Table AssistiveTouchInformation message parameters**101-38**

Name	ID	Type	#	Notes
IsEnabled	0	bool	1	

101.5.5 StopAssistiveTouchInformation

Source	ID
Accessory	0x5404

Table StopAssistiveTouchInformation message parameters**101-39**

Name	ID	Type	#	Notes
This message has no parameters.				

101.6 Bluetooth Connection Status

Requirements and usage for these messages may be found in [Bluetooth Connection Status](#) (page 316).

101.6.1 StartBluetoothConnectionUpdates

Source	ID
Accessory	0x4E03

Table StartBluetoothConnectionUpdates message parameters
101-40

Name	ID	Type	#	Notes
BluetoothTransportComponentIdentifier	0	uint16	1+	See BluetoothTransportComponent (page 876).

101.6.2 BluetoothConnectionUpdate

Source	ID
Device	0x4E04

Table BluetoothConnectionUpdate message parameters
101-41

Name	ID	Type	#	Notes
BluetoothTransportComponentIdentifier	0	uint16	1	See BluetoothTransportComponent (page 876).
ConnectedBluetoothProfiles	1	group	0/1	See BluetoothComponentProfiles (page 884).

Table BluetoothComponentProfiles parameter group
101-42

Name	ID	Type	#	Notes
BluetoothHandsFree	0	none	0/1	
BluetoothPhoneBookAccess	1	none	0/1	
BluetoothAudioVideoRemoteControl	3	none	0/1	
BluetoothAdvancedAudioDistribution	4	none	0/1	
BluetoothHumanInterfaceDevice	5	none	0/1	
BluetoothiAP2Link	7	none	0/1	
BluetoothPersonalAreaNetworkAccessPoint	8	none	0/1	Device is in the access point role.
BluetoothMessageAccess	9	none	0/1	
BluetoothPersonalAreaNetworkClient	12	none	0/1	Device is in the client role.

101.6.3 StopBluetoothConnectionUpdates

Source	ID
Accessory	0x4E05

Table 101-43 StopBluetoothConnectionUpdates message parameters

Name	ID	Type	#	Notes
This message has no parameters.				

101.7 Communications

Requirements and usage for these messages may be found in [Communications](#) (page 319).

101.7.1 StartCallStateUpdates

Source	ID
Accessory	0x4154

Table 101-44 StartCallStateUpdates message parameters

Name	ID	Type	#	Notes
RemoteID	0	none	1	
DisplayName	1	none	1	
Status	2	none	1	
Direction	3	none	1	
CallUUID	4	none	1	
AddressBookID	6	none	0/1	
Label	7	none	0/1	
Service	8	none	1	
IsConferenced	9	none	1	
ConferenceGroup	10	none	1	
DisconnectReason	11	none	0/1	
StartTimestamp	12	none	0/1	

101.7.2 CallStateUpdate

Source	ID
Device	0x4155

Table 101-45 CallStateUpdate message parameters

Name	ID	Type	#	Notes
RemoteID	0	utf8	0/1	Remote phone number or email.
DisplayName	1	utf8	0/1	Caller's display name on phone. If in Contacts: John Smith If not in Contacts: (408) 996-1010
Status	2	enum	1	See CallStateUpdateStatus (page 886).
Direction	3	enum	0/1	See CallStateUpdateDirection (page 887).
CallUUID	4	utf8	0/1	
AddressBookID	6	utf8	0/1	
Label	7	utf8	0/1	Caller's label. If in Contacts: mobile, work, home If not in Contacts: San Jose, CA
Service	8	enum	0/1	See CallStateUpdateService (page 887).
IsConferenced	9	bool	0/1	Call is part of a conference.
ConferenceGroup	10	uint8	0/1	Conference group number. Will only be sent if IsConferenced is True.
DisconnectReason	11	enum	0/1	See CallStateUpdateDisconnectReason (page 887). Will only be sent if Status = Disconnected.
StartTimestamp	12	secs64	0/1	Time interval in seconds since reference date (Jan 1, 1970, GMT). Will only be sent if the call timer has started. To determine call duration, this value should be subtracted from the current time after it has been synchronized to the DeviceTimeUpdate (page 895) message SecondsSinceReferenceDate parameter.

Table 101-46 CallStateUpdateStatus enum

Value	Description
0	Disconnected
1	Sending
2	Ringing
3	Connecting
4	Active
5	Held

Value	Description
6	Disconnecting

Table 101-47 CallStateUpdateDirection enum

Value	Description
0	Unknown
1	Incoming
2	Outgoing

Table 101-48 CallStateUpdateService enum

Value	Description
0	Unknown
1	Telephony
2	FaceTimeAudio
3	FaceTimeVideo

Table 101-49 CallStateUpdateDisconnectReason enum

Value	Description
0	No Reason (Call Ended)
1	Call Declined
2	Call Failed

Table 101-50 CallStateUpdateStatusLegacy enum

Value	Description
0	Disconnected
1	Active
2	Held
3	Ringing/Sending

Table CallStateUpdateDirectionLegacy enum
101-51

Value	Description
0	Incoming
1	Outgoing
2	Unknown

101.7.3 StopCallStateUpdates

Source	ID
Accessory	0x4156

Table StopCallStateUpdates message parameters
101-52

Name	ID	Type	#	Notes
This message has no parameters.				

101.7.4 StartCommunicationsUpdates

Source	ID
Accessory	0x4157

Table StartCommunicationsUpdates message parameters
101-53

Name	ID	Type	#	Notes
SignalStrength	0	none	0/1	
RegistrationStatus	1	none	0/1	
AirplaneModeStatus	2	none	0/1	
CarrierName	4	none	0/1	
CellularSupported	5	none	0/1	
TelephonyEnabled	6	none	0/1	
FaceTimeAudioEnabled	7	none	0/1	
FaceTimeVideoEnabled	8	none	0/1	
MuteStatus	9	none	0/1	
CurrentCallCount	10	none	0/1	
NewVoicemailCount	11	none	0/1	

Name	ID	Type	#	Notes
InitiateCallAvailable	12	none	0/1	
EndAndAcceptAvailable	13	none	0/1	
HoldAndAcceptAvailable	14	none	0/1	
SwapAvailable	15	none	0/1	
MergeAvailable	16	none	0/1	
HoldAvailable	17	none	0/1	

101.7.5 CommunicationsUpdate

Source	ID
Device	0x4158

Table 101-54 CommunicationsUpdate message parameters

Name	ID	Type	#	Notes
SignalStrength	0	enum	0/1	See CommunicationsUpdateSignalStrength (page 890). Will not be sent if CellularSupported is False.
RegistrationStatus	1	enum	0/1	See CommunicationsUpdateRegistrationStatus (page 890). Will not be sent if CellularSupported is False.
AirplaneModeStatus	2	bool	0/1	
CarrierName	4	utf8	0/1	Will not be sent if CellularSupported is False.
CellularSupported	5	bool	0/1	
TelephonyEnabled	6	bool	0/1	
FaceTimeAudioEnabled	7	bool	0/1	
FaceTimeVideoEnabled	8	bool	0/1	
MuteStatus	9	bool	0/1	
CurrentCallCount	10	uint8	0/1	
NewVoicemailCount	11	uint8	0/1	
InitiateCallAvailable	12	bool	0/1	Allowed to initiate a call or add a call if one is already active.
EndAndAcceptAvailable	13	bool	0/1	
HoldAndAcceptAvailable	14	bool	0/1	
SwapAvailable	15	bool	0/1	
MergeAvailable	16	bool	0/1	

Name	ID	Type	#	Notes
HoldAvailable	17	bool	0/1	You cannot place a call on hold if this is False. You can un-hold a call if the status is held even if this is False.

Table CommunicationsUpdateSignalStrength enum
101-55

Value	Description
0	0 bars
1	1 bar
2	2 bars
3	3 bars
4	4 bars
5	5 bars

Table CommunicationsUpdateRegistrationStatus enum
101-56

Value	Description
0	Unknown
1	Not Registered
2	Searching
3	Denied
4	Registered Home
5	Registered Roaming
6	Emergency Calls Only

101.7.6 StopCommunicationsUpdates

Source	ID
Accessory	0x4159

Table StopCommunicationsUpdates message parameters
101-57

Name	ID	Type	#	Notes
This message has no parameters.				

101.7.7 InitiateCall

Source	ID
Accessory	0x415A

Table 101-58 InitiateCall message parameters

Name	ID	Type	#	Notes
Type	0	enum	1	See InitiateCallType (page 891).
DestinationID	1	utf8	0/1	Shall only be provided when Type is Destination. Number to call. Format: 4089961010 Format: +14089961010 Format: +4408442090611
Service	2	enum	0/1	Shall only be provided when Type is Destination. See InitiateCallService (page 891).
AddressBookID	3	utf8	0/1	

Table 101-59 InitiateCallType enum

Value	Description
0	Destination
1	Voicemail
2	Redial

Table 101-60 InitiateCallService enum

Value	Description
1	Telephony
2	FaceTimeAudio
3	FaceTimeVideo

101.7.8 AcceptCall

Source	ID
Accessory	0x415B

Table AcceptCall message parameters
101-61

Name	ID	Type	#	Notes
AcceptAction	0	enum	1	See AcceptCallAcceptAction (page 892).
CallUUID	1	utf8	0/1	UUID of call to accept.

Table AcceptCallAcceptAction enum
101-62

Value	Description
0	Accept/HoldAndAccept
1	EndAndAccept

101.7.9 EndCall

Source	ID
Accessory	0x415C

Table EndCall message parameters
101-63

Name	ID	Type	#	Notes
EndAction	0	enum	1	See EndCallEndAction (page 892).
CallUUID	1	utf8	0/1	UUID of call to end.

Table EndCallEndAction enum
101-64

Value	Description
0	End/Decline
1	EndAll

101.7.10 SwapCalls

Source	ID
Accessory	0x415D

Table SwapCalls message parameters
101-65

Name	ID	Type	#	Notes
This message has no parameters.				

101.7.11 MergeCalls

Source	ID
Accessory	0x415E

Table MergeCalls message parameters
101-66

Name	ID	Type	#	Notes
This message has no parameters.				

101.7.12 HoldStatusUpdate

Source	ID
Accessory	0x415F

Table HoldStatusUpdate message parameters
101-67

Name	ID	Type	#	Notes
HoldStatus	0	bool	1	
CallUUID	1	utf8	0/1	UUID of call to update hold status.

101.7.13 MuteStatusUpdate

Source	ID
Accessory	0x4160

Table MuteStatusUpdate message parameters
101-68

Name	ID	Type	#	Notes
MuteStatus	0	bool	1	

101.7.14 SendDTMF

Source	ID
Accessory	0x4161

Table SendDTMF message parameters
101-69

Name	ID	Type	#	Notes
Tone	0	enum	1	See SendDTMFTone (page 894).
CallUUID	1	utf8	0/1	UUID of call to play tone.

Table SendDTMFTone enum
101-70

Value	Description
0	Number 0
1	Number 1
2	Number 2
3	Number 3
4	Number 4
5	Number 5
6	Number 6
7	Number 7
8	Number 8
9	Number 9
10	Star (*)
11	Pound (#)

101.8 Device Notifications

Requirements and usage for these messages may be found in [Device Notifications](#) (page 323).

101.8.1 DeviceInformationUpdate

Source	ID
Device	0x4E09

Table 101-71 DeviceInformationUpdate message parameters

Name	ID	Type	#	Notes
DeviceName	0	utf8	0/1	Device name as shown in Settings (for example, "John Doe's iPhone").

101.8.2 DeviceLanguageUpdate

Source	ID
Device	0x4E0A

Table 101-72 DeviceLanguageUpdate message parameters

Name	ID	Type	#	Notes
DeviceLanguage	0	utf8	0/1	ISO 639-1 or ISO 639-2 designation for the current device language. See https://www.loc.gov/standards/iso639-2/php/English_list.php for a complete list of ISO 639-1 and ISO 639-2 codes.

101.8.3 DeviceTimeUpdate

Source	ID
Device	0x4E0B

Table 101-73 DeviceTimeUpdate message parameters

Name	ID	Type	#	Notes
SecondsSinceReferenceDate	0	secs64	1	Time interval in seconds since reference date (Jan 1, 1970, GMT).
TimeZoneOffsetMinutes	1	int16	1	Difference in minutes between the device time zone and GMT.
DaylightSavingsOffsetMinutes	2	int8	1	Daylight savings time offset in minutes.

101.8.4 DeviceUUIDUpdate

Source	ID
Device	0x4E0C

Table 101-74 DeviceUUIDUpdate message parameters

Name	ID	Type	#	Notes
UUID	0	utf8	1	This UUID will be the same over all transports as long as one iAP2 accessory is connected. The UUID will be changed no less than one minute after all iAP2 accessories disconnect.

101.9 External Accessory Protocol

Requirements and usage for these messages may be found in [External Accessory Protocol](#) (page 352).

101.9.1 StartExternalAccessoryProtocolSession

Source	ID
Device	0xEA00

Table 101-75 StartExternalAccessoryProtocolSession message parameters

Name	ID	Type	#	Notes
ExternalAccessoryProtocolIdentifier	0	uint8	1	See ExternalAccessoryProtocol (page 874).
ExternalAccessoryProtocolSessionIdentifier	1	uint16	1	

101.9.2 StopExternalAccessoryProtocolSession

Source	ID
Device	0xEA01

Table 101-76 StopExternalAccessoryProtocolSession message parameters

Name	ID	Type	#	Notes
ExternalAccessoryProtocolSessionIdentifier	0	uint16	1	